

CO2 AT1

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Course Name: Computer Vision

Grey Level Transformation

- a) Grey level transformation is a point processing techniques used to modify the brightness & contrast of an image. It changes the intensity value of each pixel independently to improve quality, highlight important features.

Applications :

- Image enhancement
- Medical imaging
- Satellite image analysis
- Object detection

- b) Apply negative transformation

$$S = 255 - r$$

Original Pixel (r)

50

100

150

200

Transformed Pixel (S)

$$255 - 50 = 205$$

$$255 - 100 = 155$$

$$255 - 150 = 105$$

$$255 - 200 = 55$$

Final Answer $\Rightarrow [205, 155, 105, 55]$

2) Log Transformation

- a) Log transformation expands low-intensity (dark) pixel values while compressing high-intensity values. This makes in dark regions more visible.

$$S = c \log(1 + r)$$

where

c = Constant

r = input pixel

b) $c = 10$

using natural logarithm ($\ln$)

for $r = 0$

$$S = 10 \ln(1) = 0$$

$r = 10$

$$S = 10 \ln(11)$$

$$= 10 (2.3979) = 23.98$$

$r = 50$

$$S = 10 \ln(51)$$

$$= 10 (3.9318) = 39.32$$

$r = 100$

$$S = 10 \ln(101)$$

$$= 10 (4.6151) = 46.15$$

Final Answer

$$\Rightarrow [0, 23.98, 39.32, 46.15]$$

Histogram Processing

- a) Histogram equalization redistributes the intensity values of an image to improve contrast. It spreads frequently occurring intensity values over the full intensity range, making details more visible.

Advantages:

- Improves image contrast
- Enhances hidden details
- Makes dark images brighter.

- b) Compute CDF & Equalized values.

Given Histogram [3bit image]

Intensity	Frequency
0	2
1	2
2	4
3	8

Total Pixels

$$\Rightarrow 2+2+4+8$$

$$= 16$$

Step-I

Intensity	Frequency	CDF
0	2	2
1	2	2
2	4	8
3	8	16

Step-2 Equalized Intensity

For 3-bit image

$$L = 8$$

$$s = \frac{CDF}{N} [L-1]$$

Since $L-1 = 7$

Intensity 0 $\Rightarrow \frac{2}{16} \times 7 = 0.875 \approx 1$

Intensity 1 $\Rightarrow \frac{2}{16} \times 7 = 1.75 \approx 2$

Intensity 2 $\Rightarrow \frac{8}{16} \times 7 = 3.5 \approx 4$

Intensity 3 $\Rightarrow \frac{16}{16} \times 7 = 7$

Final Answers

Equalized.

0	1
1	2
2	4
3	7

4) Spatial Filtering [Smoothing]

- a) Spatial smoothing reduces image noise & small variations by averaging neighboring pixel values.

Applications:

- Noise Reduction
- Image Preprocessing
- Preparing images for segmentation

b) Apply 3×3 averaging filter

Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Sum of Pixels

$$\Rightarrow 10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50$$

$$= 270$$

$$\text{Average} = \frac{270}{9} = 30$$

$$\therefore \text{Filter} = 30$$

5) Spatial Filtering

a) Sharpening emphasizes edges & fine details by increasing intensity differences between neighboring pixels. It highlights boundaries & improves image clarity.

b) Apply Laplacian Mask

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Center Calculation

$$(4 \times 20) - (10 + 10 + 10)$$

$$80 - 40 = 40$$

$$= 40$$

6) Frequency Domain Filtering

a) Differentiate

Low-pass filter

Keeps low frequencies

Removes noise

Produces smooth image

Blurs image

High Pass filter

Keeps high frequencies

Enhances edges

Produces sharp image

Highlights details.

b) Apply low pass filter.

Given $F = [100, 80, 40, 20]$

Keep first two freq

Filtered output $\rightarrow [100, 80, 0, 0]$

Thresholding

- a) Global Thresholding uses a single threshold value for the entire image

Rule

If

$$\text{Pixel} > T$$

$$\text{Output} = 1$$

Else

$$\text{Output} = 0$$

It is simple and commonly used for image segmentation.

- b) Pixels [200]

$$\text{Threshold } T = 100$$

$$\text{Since } 200 > 100$$

$$\text{output} = [1]$$

8) Edge Detection

Edge detection identifies sudden changes in image intensity

It is used to detect boundaries and shapes.

Applications:

- Object detection
- Face recognition
- Medical Imaging
- Computer Vision.

b) Horizontal Sobel operator.

$$\text{Kernel} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{Image} = \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$\text{First row} = (-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -40$$

$$\text{Middle row} = 0$$

$$\text{Bottom row} = (1 \times 30) + (2 \times 30) + (1 \times 30) = 120$$

$$\begin{aligned} \text{Gradient} &\Rightarrow 120 - 40 \\ &= 80 \end{aligned}$$

9) Region Based Segmentation.

a) Region growing starts from a seed pixel & includes neighbouring pixels whose intensity difference from the seed is within specified threshold.

Advantages:

- Produces connected regions
- suitable for medical images
- Easy to implement.

b) Identify Region.

The question appears to contain a typo. It lists pixel intensities as $[1, 0, 2]$ but then states seed = 50 and threshold difference = 5.

These values are inconsistent.

If intended pixel values were $[48, 50, 52]$

Seed = 50

Threshold = 5

Differences:

Pixel	Differences
48	2
50	0
52	2

All differences are ≤ 5

$[48, 50, 52]$ all belong to the region.

10) Combined Enhancement & Segmentation.

- a) Smoothing removes image noise before segmentation. This reduces false edges & produces more accurate segmented regions.

Benefits:

- Removes the noise

- Improves thresholding accuracy
- Gives cleaner segmented objects

b) Apply averaging & threshold.

Given

Image values [10, 50, 85]

Assume Average 48

Threshold $T = 50$

Rule

if Pixel $\geq 50 \rightarrow 1$

Else $\rightarrow 0$

average value = 48 $\nless$ 50

Segmented output = 0,,